



ST BENEDICT'S

SUBJECT	PHYSICS FINAL EXAMINATION
GRADE	11
EXAMINER	Mr McCoy
MODERATOR	Mrs Govender
DATE	09 November 2023
MARKS	200
DURATION	3 hours

%

LEARNER NAME:

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 11 pages, including additional space. A Data Sheet has been provided. Read the questions carefully.
2. Answer **ALL** the questions on the question paper and hand Units need not be included in the working of calculations, but appropriate units should be

QUESTION 1 MULTIPLE CHOICE

14 MARKS

1.1 B✓✓

1.2 B✓✓

1.3 A✓✓

1.4 B✓✓

1.5 B✓✓

1.6 A✓✓

1.7 B✓✓

1.8 D✓✓

1.9 B✓✓

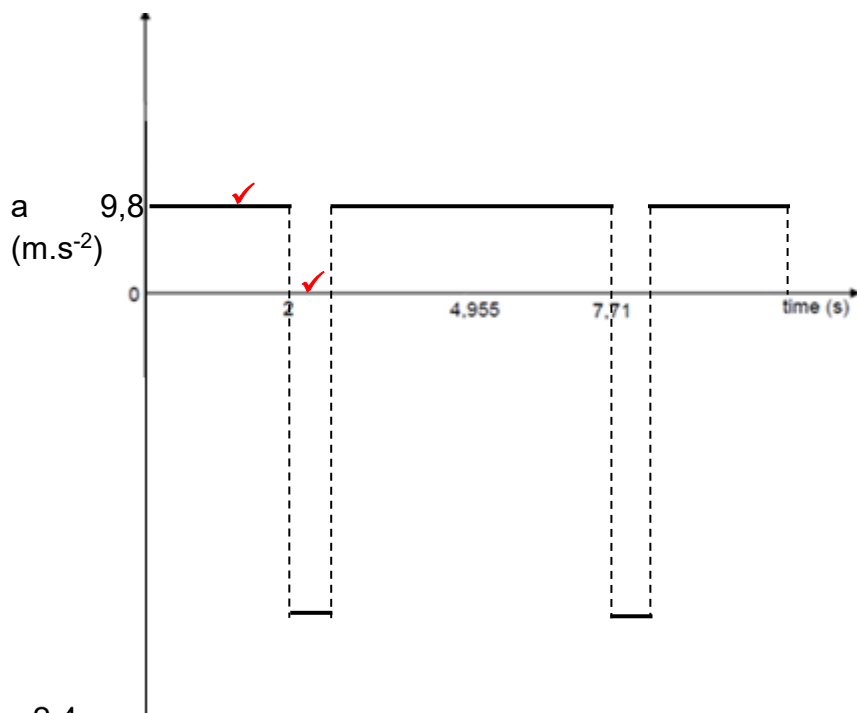
1.10 B✓✓

QUESTION 2

2.1 The rate of change of position **or** the rate of displacement✓✓ **or** the rate of change of displacement (2)

2.2 $10 \text{ m}\cdot\text{s}^{-1}$ ✓ (1)

2.3 The gradient represents the acceleration due to gravity (g) which is constant for free fall. ✓✓ (2)



Comment: dotted line not drawn below zero line

2.4

2.5

$$2.5.1 \quad s = ut + \frac{1}{2}at^2 \quad \checkmark \quad \text{or } s = \frac{(u+v)t}{2} \quad t = \frac{(29.6+10)}{2} = 39.6 \text{ m}$$

$$= (10)(2) + \frac{1}{2}(9,8)(22) \quad \checkmark$$

$$= 39,6 \text{ m} \quad \checkmark \quad (3)$$

OR

$$\begin{aligned} \text{Height} &= \text{Area under the graph} \\ &= \text{Area of } \square + \text{Area of } \triangle \quad \checkmark \\ &= (10)(2) + \left(\frac{1}{2}\right)(2)(19,6) \quad \checkmark \\ &= 39,6 \text{ m} \quad \checkmark \end{aligned}$$

$$2.5.2 \quad v = u + at \quad \checkmark \quad (4)$$

$$0 = -25 \quad \checkmark + (9,8)t \quad \checkmark$$

$$t = 2,55 \text{ s}$$

$$\text{Total time } T = 8 + 2,55 = 10,55$$

2.6

$$2.6.1 \quad 2 \text{ s} \quad \checkmark \quad (1)$$

$$2.6.2 \quad 4,955 \text{ s} \quad \checkmark \quad (1)$$

$$2.6.3 \quad -27 \text{ (m}\cdot\text{s}^{-1}) \quad \checkmark \quad \text{Comment: negative often missing} \quad (1)$$

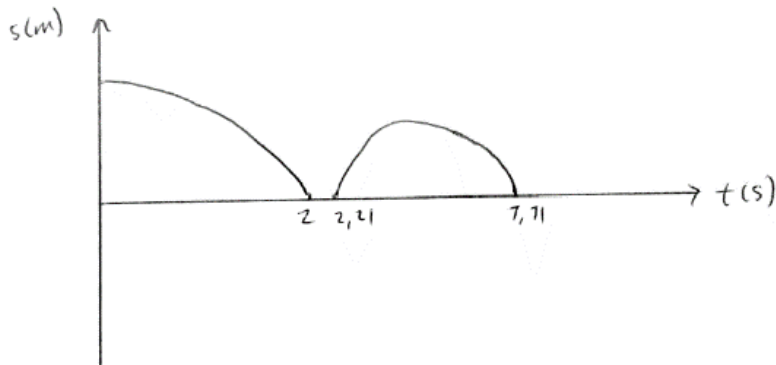
2.7 Down as +: Up as +:

✓ curve up to 2 s

✓ graph between 2,2s and 7,71s

✓ gap **and** height loss (3)

Comment: gap often missing



QUESTION 3**20 MARKS**

3.1 .

3.1.1 Speed is the rate of the change of distance. ✓✓ (2)

3.1.2 $v = \frac{s}{t} = \frac{150}{17} \checkmark = 8,82 \text{ m.s}^{-1} \checkmark$ (3)

3.2 .

3.2.1 **Resultant vector** is the single vector which has the same effect as the original vectors acting together. 2.2.2 (2)

3.2.2 $v = (4,9 - 0,5) = 4,4 \text{ m.s}^{-1} \checkmark$ **West** ✓ not left (2)

3.2.3 Kuhle: $t = \frac{s}{v} = \frac{420}{4,4} = 95,45 \text{ s} \checkmark \checkmark$

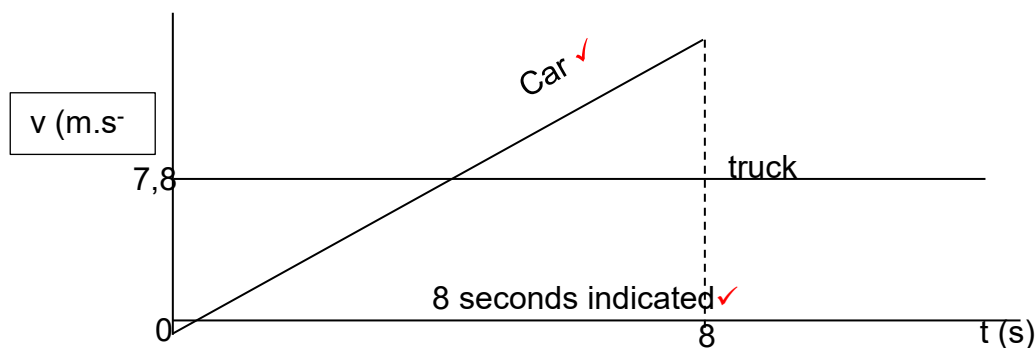
Pili: $t = \frac{s}{v} = \frac{550}{4,8} = 114,58 \text{ s} \checkmark \checkmark$

Kuhle reaches the jetty first ✓ (5)

3.3 .

3.3.1 $v = u + at \checkmark$
 $= 0 + 4,2 (8) \checkmark$
 $= 33,6 \text{ m.s}^{-1} \checkmark$ (3)

3.3.2



3.3.3 .

$s_{\text{CAR}} = ut + \frac{1}{2} at^2$ $= 0(8) + \frac{1}{2} (4,2) 8^2 \checkmark$ $= 134,4 \text{ m} \checkmark$	$s_{\text{TRUCK}} = ut$ $= 7,8 \times 8 \checkmark$ $= 62,4 \text{ m} \checkmark$ Distance = $134,4 - 62,4 = 72 \text{ m} \checkmark$ (5)
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3.3.4 . (4)

P	Q	join
u: 0 v: a: P t: t s: 20	u: 0 v: a: Q t: t s:	$t = t \checkmark = \sqrt{\frac{2s}{P}}$ $\therefore v_Q = Q \sqrt{\frac{2s}{P}} \checkmark$
$s = \frac{1}{2} Pt^2 \checkmark$	$v = Qt \checkmark$	

QUESTION 4

4.1 Upwards✓ (1)

4.2 B (0,32) – D (0,99s) ✓ (1)

4.3 $\Delta s = \frac{1}{2} b \times h = 0,5 \times 0,32 \times 1,375 = 0,22 \text{ m}$ ✓ (3)

4.4 $v = u + at$
 $0 = 1,372 + (-9,8) \times t$
 $\therefore t = 0,14$
 $\therefore t = 0,14 + 0,32 = 0,46 \text{ s}$ ✓ (3)

4.5 $v = u + at$
 $v = 1,372 + (-9,8)(0,67)$
 $v = 5,19 \text{ m.s}^{-1}$ ✓

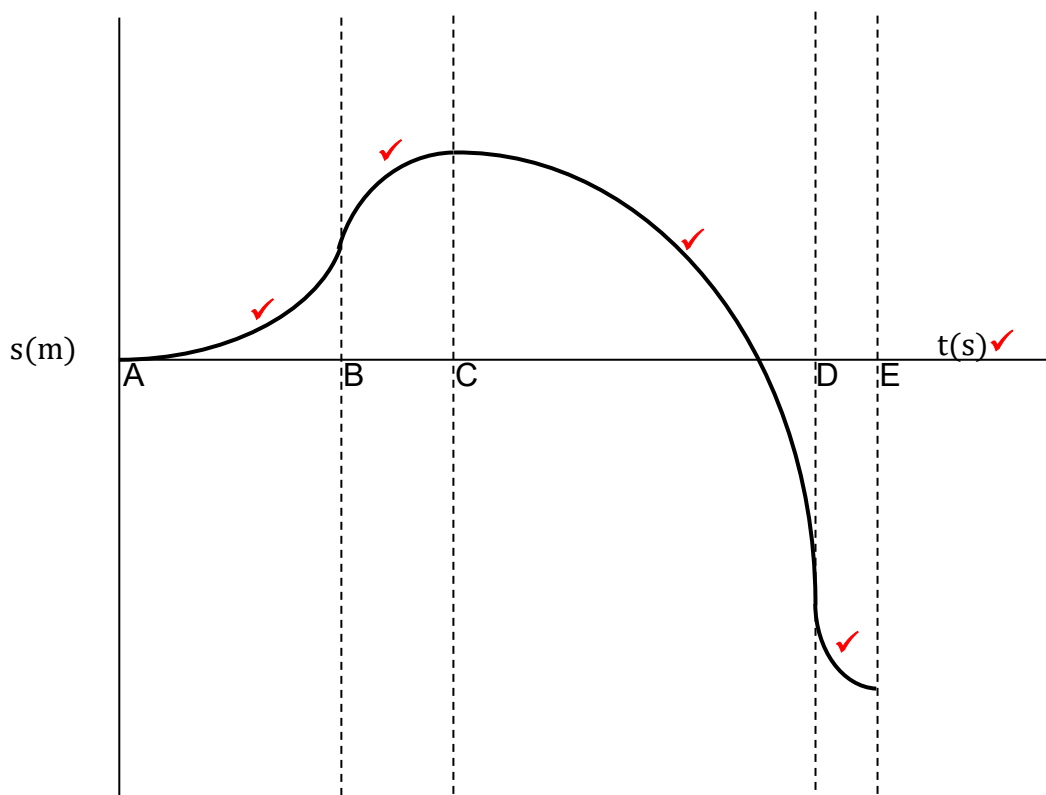
$$v^2 = u^2 + 2as$$

$$(-5,19)^2 = (1,372)^2 + 2(-9,8)s$$

$$\therefore s = -1,278 \text{ m}$$
✓

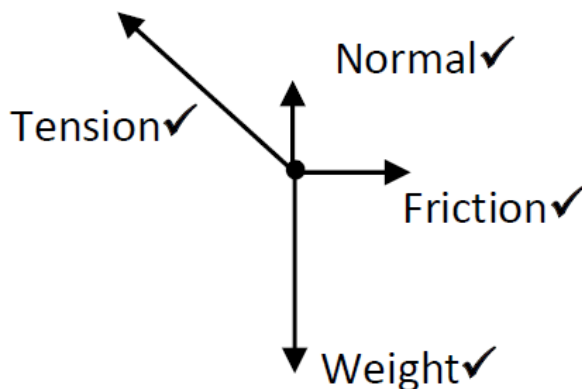
$$\therefore \text{height} = +0,22 - 1,278 = -1,06 \text{ m} \therefore 1,06 \text{ m}$$
✓ (5)

4.6 (5)



QUESTION 5**19 MARKS**

5.1 . (4)



5.2 The force that opposes the motion of an object✓✓ (2)

5.3 $F_f = \mu F_N$

$120 = 0,34 F_N$ ✓

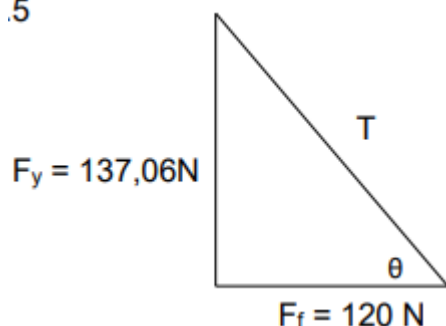
$F_N = 352,94 \text{ N}$ ✓ (2)

5.4 $N + F_y = F_g$ ✓

$352,94 + F_y = 50(9,8)$ ✓

$F_y = 137,06 \text{ N}$ (3)

.5



$\tan \theta = \frac{137,06}{120}$ ✓✓

$\theta = 48,8^\circ$ ✓

Or $T \sin \theta = 137,06 \text{ N}$

$T \cos \theta = 120 \text{ N}$

5.5 (3)

5.6 $T^2 = \sqrt{120^2 + 137,06^2}$ ✓

$T = 182,17 \text{ N}$ ✓ (2)

5.7 The external force is on the role.

The inertia of the parcel is too great to accelerate it with the rope✓.

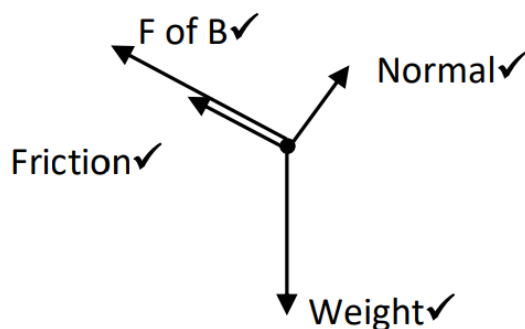
The parcel therefore pulls back on the rope✓, increasing the force✓, causing it to snap.

(3)

QUESTION 6**21 MARKS**

6.1 the gravitational force the Earth✓ exerts on any object on or near its surface✓ (2)

6.2 . (4)



Comment: friction drawn down slope

6.3 $F_f = \mu F_N$ ✓
 $= 0,52 \times (0,4)(9,8)$ ✓ $\cos 35^\circ$ ✓
 $F_f = 1,67 \text{ N}$ ✓ up the slope ✓ *Comment: direction not given* (5)

6.4 Steel makes greater contact between other surfaces than copper
 $\therefore$ it has a greater μ_s ✓ ✓ *Comment: two different surfaces not identified* (2)

6.5 $\mu_k < \mu_s$ ✓ ✓ or $F_K < F_s$ *Comment: poorly answered* (2)

6.6 $F_{\text{net}} = ma$ ✓
 $F_{g\parallel} - F_f - F_B = ma$
 $0,4(9,8)\sin 35^\circ - 0,9$ ✓ $-F_B = 0,4(3,03)$ ✓
 $2,25 - 0,9 - F_B = 1,21$
 $F_B = 0,14 \text{ N}$ ✓ up the slope ✓ *Comment: poorly arithmetic* (6)

QUESTION 7 WORK, ENERGY AND POWER**30 MARKS**

7.1 .

7.1.1 In the absence of air resistance or any external forces✓, the mechanical energy of an object is constant✓ (2)

7.1.2 $E_{m \text{ top}} = E_{m \text{ bottom}}$ ✓
 $mgh = \frac{1}{2}mv^2$
 $9,8(5) = \frac{1}{2}v^2$ ✓
 $v = 9,9 \text{ m.s}^{-1}$ ✓ (3)

7.1.3 According to NI✓ an object will continue in its state of motion✓ since no net force is applied to it✓
 Or E_m ✓ is conserved✓ in an isolated system✓ (3)

7.1.4 Normal✓✓ (Not F_g , on a slope, accept F_g perpendicular.) (2)

7.1.5 $E_{m \text{ bottom}} = E_{m \text{ top}} + W_f$ ✓
 $\frac{1}{2}mv^2 = mgh + W_f$
 $\frac{1}{2}(5)(9,9)^2 = 5(9,8)(3) + W_f$ ✓
 $W_f = 245,03 - 147$
 $= 98,03 \text{ J}$ ✓ (4)

7.1.6 the work done by a net force on an object✓ is equal to the change in the kinetic energy of the object✓ (2)

7.1.7 $\Delta E_p + W_f = 0 - \frac{1}{2}mu^2$ ✓
 $-147 + F_f\left(\frac{h}{\sin \theta}\right) = 0 - 245,02$
 $-147 + \left(\frac{-54}{\sin \theta}\right) = 0 - 245,02$ ✓
 $\theta = 33,45^\circ$ ✓ (5)

7.2 .

7.2.1 The rate at which work is done or the rate at which energy is transferred (2)

7.2.2 $P = \frac{F \times s}{t}$
 $P = \frac{150 \times 10}{8}$ ✓
 $P = 187,5 \text{ W}$ ✓ (4)

7.2.3 $\% \text{ efficiency} = \frac{P_{out}}{P_{in}} \times 100$
 $= \frac{187,5}{210} \times 100$
 $= 89,29\%$ ✓ (3)

QUESTION 8

8.1 .

8.1.1 $E = \frac{hc}{\lambda}$

$$E = \frac{6,6 \times 10^{-34} \times 3 \times 10^8}{492,1 \times 10^{-9}}$$

$$E = 4,02 \times 10^{-19} \text{ J} \quad (3)$$

8.1.2 When an electron moves from the excited state to ground state, light is emitted. (3)

8.1.3 4 (2)

8.2 Each element has a unique number of electrons.

These electrons fall to lower energy levels giving out a unique combination of frequencies/wavelengths (therefore each has a unique spectrum). (2)

QUESTION 9

9.1 State *Ohm's Law*. (2)

Current through a conductor is directly proportional ✓ to the potential difference across
the conductor at constant temperature. ✓

9.2 Determine the reading on A_1 . (2)

$$I = \frac{V}{R}$$

$$I = \frac{6,4}{4} \text{ ✓ (f and sub) }$$

$I = 1,6 \text{ A} \text{ ✓}$

9.3 1,6A ✓✓ (2)

9.4 Determine the amount of work done by R_1 in 10 minutes. (4)

$$W = I^2 R t \text{ ✓}$$

$$W = (1,6^2)(8) \text{ ✓ } (10 \times 60) \text{ ✓}$$

$W = 12\,288 \text{ J} \text{ ✓}$

9.5 Calculate the reading on voltmeter V_1 . (4)

$$emf = IR$$

$$emf = (1,6)(4 + 3 + 8) \text{ ✓}$$

$$emf = 24 \text{ V} \text{ ✓}$$

$$24 = 2V_{batt} \text{ ✓}$$

$V_{batt} = 12 \text{ V} \text{ ✓}$

9.6 Define *resistance*. (2)

The opposition to the flow of current ✓✓

9.7 : **INCREASE** ✓ (1)

9.8 Explain your answer to QUESTION 9.7 (3)

Emf remains constant ✓
Resistance will decrease due to removal of resistor in series ✓
Current is inversely proportional to resistance.
$I \propto \frac{1}{R} \text{ ✓}$

QUESTION 10

10.1 Determine the resistance of R_2 . (2)

$$R = \frac{V}{I}$$

$$R = \frac{4,5}{0,75} \checkmark$$

$$\boxed{R = 6\Omega} \checkmark$$

10.2 How much power does the cell supply to the circuit? (3)

$$P = I V_{emf} = (I)(V_{R_2} + IR_3)$$

$$P = (0,75) \checkmark (4,5 + (0,75)(2) \checkmark) = (0,75)(6)$$

$$\boxed{P = 4,5 W} \checkmark$$

Switch S_2 is now closed

10.3 What is the reading on V_1 . (1)

6V $\checkmark$

10.4 Determine the effective resistance of the circuit. (4)

$$\frac{1}{R_t} = \frac{1}{R_1} + \frac{1}{R_2 + R_3} \checkmark$$

$$\frac{1}{R_t} = \frac{1}{4} + \frac{1}{(6 + 2) \checkmark}$$

$$\frac{1}{R_t} = \frac{2}{8} + \frac{1}{8} = \frac{3}{8}$$

$$R = \frac{8}{3} = \boxed{2,67\Omega} \checkmark \checkmark$$

Switch S_2 remains closed and Switch S_3 is opened.

10.5 How will the reading on voltmeter change INCREASE, DECREASE or **REMAIN THE SAME?** $\checkmark$ Circle the correct option. (1)

10.6 Explain your answer to QUESTION 10.5. (2)

The resistors are in parallel so the voltage over them will remain the same. $\checkmark \checkmark$

END OF ASSESSMENT